

MUHAMMAD MOIZ BUKHARI

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SUMMARY

Software and systems engineer who owns problems end-to-end – from embedded firmware architecture and hardware bring-up to backend services and formal V&V in FDA-regulated environments. Built and shipped a 5-fixture automation system from the ground up at Abbott Laboratories, cutting a 72-hour test process down to zero manual intervention. Operates fluently across the stack: low-level C/C++, Python automation, SQL-backed services, and regulatory documentation.

EXPERIENCE

Systems Engineering Co-op – Abbott Laboratories

Jan 2025 – Aug 2025

Pleasanton, CA

- **Owned design and delivery** of a 5-fixture Power Automation system end-to-end, combining embedded Arduino C/C++ firmware with Python test sequencing to run 72-hour unattended endurance cycles with zero manual intervention.
- Architected modular firmware control logic that decoupled sensor drivers, relay/actuator control, and communication layers across 5 fixture revisions, establishing a reusable pattern that cut rework on every subsequent build.
- Engineered embedded control systems in Arduino C/C++ – relay boards, color sensors, LCD status displays, and custom battery-detection algorithms – taking automated power-cycling from concept to production-ready.
- Took fixture hardware from design to production single-handedly: soldering, wiring, crimping, and continuity/voltage validation against schematics, delivering builds ready for Long-Term Testing without rework cycles.
- **Drove FDA-compliant V&V execution** on the HeartMate Touch Monitor, independently running formal test protocols across multiple BLE pairing scenarios to verify system-requirement compliance.
- Set the documentation standard for the fixture program by authoring IQ/OQ validation packages and CO/CA records in Windchill PLM, ensuring full audit traceability under FDA 21 CFR 820 and ISO 13485.
- Built the test infrastructure the team relied on: Python-based sequencing, real-time logging, and anomaly detection across 4+ parallel fixture configurations, becoming the go-to engineer for cross-team debug and root-cause resolution.

Software Engineer – TWM Wholesale

May 2026 – Present

Stockton, CA

- **Eliminated 3–4 hours/day of manual data entry** by independently designing and deploying 10+ Python automation scripts across inventory and procurement workflows.
- Raised automation uptime across the operation by tracing recurring failure points through log analysis and rebuilding scripts as fault-tolerant pipelines.

PROJECTS

Power Automation Fixture – Abbott Co-op

2025

- Designed the full automation and status-reporting layer for parallel fixture builds – signal processing, custom battery-detection algorithms, and relay control logic – from a blank page.
- Drove firmware reliability higher by debugging state-machine logic via serial monitors and live LCD diagnostics, iterating fast on hardware-in-the-loop behavior until cycles ran clean.

WaterCon – Cloud-Connected IoT Platform

Personal Project

- Took stakeholder reporting from hours to seconds by designing and building RESTful APIs and a normalized SQL schema for real-time telemetry across 3+ sensor device types.
- Delivered 99.5% platform uptime through fault-tolerant Python ingestion pipelines with automated anomaly detection and alerting, built solo end-to-end.

End-to-End Encrypted Messaging App

Personal Project

- Built enterprise-grade message security from scratch – custom AES-256 encryption and Diffie-Hellman key exchange – with zero external dependencies.
- Engineered real-time concurrent delivery via a multithreaded client-server architecture with persistent session management over raw sockets.

SKILLS

Core Languages: Python, C/C++, SQL, Bash, PowerShell, MATLAB

Backend & Data: RESTful API Design, Relational DB Design, Query Optimization, ETL Pipelines

Embedded & Systems: Arduino/AVR, UART/SPI/I2C, GPIO, ADC/PWM, Relay Control, Firmware Architecture

Platforms & Tools: Linux (Ubuntu, CentOS), Git/BitBucket, VMware, Windchill PLM

Regulated & Quality: FDA 21 CFR 820, ISO 13485, IEC 62304, V&V Protocols, IQ/OQ/PQ Validation

Hardware & Lab: Soldering, Wire Stripping/Crimping, Continuity/Voltage Testing, Schematic Reading

Additional: Java, JavaScript, PHP • Agile/Scrum (Certified Scrum Master), Cross-Functional Collaboration, Root-Cause Analysis, Technical Documentation

EDUCATION

B.S. Computer Science, Honors – University of the Pacific, Stockton, CA | Aug 2021 – Dec 2025 | GPA: 3.83 | Dean's List (All Semesters)